Computer Science & IT



C Programming

Control Flow Statement

Lecture No. 01



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Recap of Previous Lecture



- Topic — shift left
- Topic — shift right
- Topic — Scope of variable
- Topic — printf return value
- Topic

Slide

Topics to be Covered



- Topic
- Topic
- Topic
- Topic
- Topic

Ternary operator

if-else

for loop

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Associativity Rule

Precedence	Operator	Description	Associativity
1	`++ --`	Suffix/postfix increment and decrement	Left-to-right
	`()`	Function call	
	`[]`	Array subscripting	
	`.`	Structure and union member access	
	`->`	Structure and union member access through pointer	
	`(type){list}`	Compound literal(C99)	
2	`++ --`	Prefix increment and decrement	Right-to-left
	`+ -`	Unary plus and minus	
	`! ~`	Logical NOT and bitwise NOT	
	`(type)`	Type cast	
	`*`	Indirection (dereference)	
	`&`	Address-of	
	`sizeof`	Size-of	
	`_Alignof`	Alignment requirement(C11)	

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Associativity Rule

3	* / %	Multiplication, division, and remainder	Left-to-right
4	+ −	Addition and subtraction	
5	<< >>	Bitwise left shift and right shift	
6	< <= > >=	For relational operators < and ≤ respectively For relational operators > and ≥ respectively	
7	== !=	For relational = and ≠ respectively	
8	&	Bitwise AND	
9	^	Bitwise XOR (exclusive or)	
10	|	Bitwise OR (inclusive or)	
11	&&	Logical AND	
12	||	Logical OR	



Associativity Rule

13	?:	Ternary conditional	Right-to-Left
14	= += -= *= /= %= <<= >>= &= ^= |=	Simple assignment Assignment by sum and difference Assignment by product, quotient, and remainder Assignment by bitwise left shift and right shift Assignment by bitwise AND, XOR, and OR	
15	,	Comma	Left-to-right



Bit-wise Operator

```
#include <stdio.h>
    int main ()  {
      int x = 5,y=36, z;
          z = ~x + y >> 2;
       printf("%d", z);
    }
```

(A) 0

(B) 7

(C) 12

(D) -6

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Bit-wise Operator

```
#include <stdio.h>
    int main ()  {
      int x = 5,y=24, z;
          z = x | y ^ 10;
       printf("%d", z);
    }
```

(A) 23

(B) 4

(C) 12

(D) -6



Print() Return value

```
#include<stdio.h>
int main(){
   printf("%d",printf("ABCD"));
   return 0;} (
```

(A) ABCD4

(B) 4

(C) ABCD 4

(D) ABCD

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Print() Return value

```
#include <stdio.h>
  int main(){
     printf("\n%d", printf("ABCD"));
     return 0;
   }
```

Slide



Print() Return value

```
#include <stdio.h>
  int main(){
     printf("\n%d", printf("ABCD"));
     return 0;
   }
```

```
ABCD
4
```

Slide



Print() Return value

```
#include <stdio.h>
  int main(){
     int a=1000;
     printf("%d",printf("\n%d",a));
     return 0;
}
```

Output of the program is _____

(A) 1005

(B) 10005

(C) 1000

(D) 1000
    5

int

char

float data

float Size : 4B
32bits

Double Size : 8Byte 64bits

```
float pi = 3.14
printf("%f", pi);
```

```
Double pi = 314
printf("%lf", pi);
```

IEEE-754 ← Important COA

1. Single precision - 32bit
2. Double precision - 64bits

* Modulo operator does not apply Not float



Ternary Operator

Conditional operator

expression 1 ? expression 2 : expression 3

↑ Relational expression

if expression 1 is (true / Nonzero)
expression 2 will execute

expression 1 is (zero / false)
expression 3 will execute.



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Ternary Operator

```
#include <stdio.h>
int main()
{
    int a;
    a = 10>7?10:20;
    printf("%d", a);
    return 0;
}
```

(A) 10
(B) 20
(C) 1
(D) 0

a : 10>7 ? 10:20

a : 10

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Ternary Operator

```
#include<stdio.h>
int main(){
      int x=3, y=4, z=4;
      printf("%d", (z>=y>=x?100:200));
      return 0 ;
}
```

(a) 100 (b) 200 (c) 0 (d) 1

[B]

z>=y>=x

4>=4>=3

1>=3 false

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Ternary Operator

```
#include <stdio.h>

int main(){
    int a = 10, b=14;

    a == 4 ? printf("%d",a):printf("%d", b);

    return 0;
}
```

a == 4

10 == 4 - false

(A) 10 (B) 14 (C) 4 (D) Error

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Ternary Operator

```
#include <stdio.h>

int main(){
    int a = 10, b=14;

    (a = 4) ? printf("%d",a):printf("%d", b);

    return 0;
}
```

(A) 10 (B) 14 (C) 4 (D) Error

(a=4)

a 10 → 4

a Nonzero — Hence it's true

4

a is already update

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Ternary Operator

= 4 ? ? :

```
#include <stdio.h>

int main(){
    int a = 10, b=14;

    a = 4 ? printf("%d",a):printf("%d", b);

    return 0;
}
```

4 is Nonzero value means its a true Statem

(A) 10 (B) 14 (C) 4 (D) Error

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Ternary Operator

10

```
#include <stdio.h>

int main(){
    int a = 10, b=14;

    a = 4 ? printf("%d",a):printf("%d", b);
    return 0;
}
```

a = 4? 2 : printf("%d", b);

a = 2

printf return No. of character printed

printf("%d", a); ⇒ this will print (2)

(A) 10 (B) 14 (C) 4 (D) Error

Slide



Ternary Operator

```
#include <stdio.h>

int main(){
    int a = 10, b=14;

    a = 4 ? printf("%d",a):printf("%d", b);
    printf(“%d”,a);
    return 0;
}
```

(A) 102 (B) 104 (C) 142 (D) 144



Control Flow Statement

Sequential Flow of Execution:

```
int main() {
    S1;
    S2;
    S3;
    S4
    S5
    S6
    S7;
}
```

processor executes the statements in Sequential manner one after another.

Can we change Sequential flow execution

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Types of Control Flow statement

- Selection or Branching or Decision or Conditional ← if-else
- Iterative statement — loops . for, while, do while
- Jump statements → break, continue



if statement syntax

```
if (expression) {
}
```

Relational Expression

0 — zero

1 — No zero

if relational expression is Not present then we evaluate Nonzero or zero

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if statement Example

```
#include <stdio.h>
int main (){
        int a = 10;
        if( a < 20) {
            printf("a is less than 20\n" );
        }
        printf("value of a is : %d\n", a);
        return 0;
}
```

will execute only condition is true

a is less than 20

value of a is 10

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if statement Example

```
#include <stdio.h>
      int main (){
            int a = 10;
            if(a+2) {
                  printf("Namaskar" );
            }
            printf("Dosto");
            return 0;
}
```



if statement Example

```
#include <stdio.h>
int main (){
    int a = 10;
    if(a-10) {
        printf("Namaskar" );
    }
    printf("Dosto");
    return 0;
}
```

if block is Not parent

Dosto

Single statement does not required blocking

if(10-10)

if(0)

if(a-10)
    printf("Namaskar");

Same output : Dosto

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if statement Example

```
#include <stdio.h>
        int main (){
                int a;
                if(a=10) {
                        printf(“Namaskar" );
                }
                printf(“Dosto");
                return 0;
}
```



Question

```
#include <stdio.h>
int main (){

      int i=0;
      if(i++){
            printf("Namaskar");
      }
      printf("\t Dunia");
      return 0;
}
```

(A) NamaskarDunia

(B) Namaskar Dunia

(C) Namaskar

(D) Dunia

if(i++) → post increment

, i++

Same

if(++i)

if(0)

Slide

```c
#include <stdio.h>
int main (){

        int a=0;
        if(a++);
        {
                a = a+10;
                a = a-5;

        }
        printf("%d", a);
        return 0;
}
```

a 1

Semicolon

if statement followed by semicolon terminates if statement

a : 1 1

a : 11-5 = 6

(A) 5 (B) 6 (C) 15 (D) 20

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If else Statement

```
int main(){
        s1;
        if(condition/expression){
                s2;
                s3;
        }
        else {
                s4;
                s5;

        }
        s6;
}
```

true

false

true

jump - branch



Question

```
#include <stdio.h>
     int main (){

          int i=1;
          if(i+2-3){
               printf("Hello friends");
          }
          else
               printf("mai bhi hu");
          return 0;
}
```

(a) Hello friends
(b) mai bhi hu
(c) Hello frinds mai bhi hu
(d) No Output

(1+2-3)



Question

```
#include <stdio.h>
     int main (){

          int i=0;
          if(i+2-3){
               printf("Hello friends");
          }
          else
               printf("mai bhi hu");
          return 0;
}
```

(a) Hello friends
(b) mai bhi hu
(c) Hello frinds mai bhi hu
(d) No Output

if(-1) → Nonzero

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```
# include<stdio.h>
int main() {
    int a;
    a = 4 ? printf("10") ? 10 : 20 : 30;
    printf ("%d", a);
}
```

Right Associative

2

a = 4 ? 10 : 30

a : 10

10

```
# include<stdio.h>
int main() {

    int a;

    a = 10;

    printf("%d", a);
}
```

Right Associative

1010



If else if Statement

```
#include <stdio.h>
int main(){
      int number = 9;
      if(number==10){
         printf("%d number is equal to 10",number);
      }
      else if (number<10) {
         printf("%d is less than 10", number);
      }
     else {
         printf("%d is greater than 10", number);
     }

     return 0;
} Number of print statement executed is ____
```

9==10 ← false

9 < 10

9 is less than 10

X

1

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Nested if else Statement

```
#include <stdio.h>
int main() {
    int i = 10;
    if (i == 10) {
        if (i < 15)
            printf("i is smaller than 15\n");

        if (i < 12)
            printf("i is smaller than 12 too\n");
        else
            printf("i is greater than 15");
    }

    return 0;
}
```

10 == 10 ← true

10 < 15

10 < 12

10 < 12

if statement block contain another if statement

Number of print statement executed is 2

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If else ladder

```
#include <stdio.h>
int main() {
      int i = 20;

      if (i == 10)
            printf("i is 10");
      else if (i == 15)
            printf("i is 15");
      else if (i == 20)
            printf("i is 20");
      else
            printf("i is not present");
}
```



2 mins Summary

Topic — Ternary operator

Topic — if statement

Topic — if-else statement

Topic

Topic

THANK - YOU

